

DEGREE IN SCIENCE		
Programme: <i>Degree in Science</i>		Year: III Semester: V Paper-I
Subject: Physics		
Course Code:	Course Title: Physical Optics	
Course Outcomes: 1. Study of Interference of light. Interference by division of wavefront and division of amplitude. 2. Understanding Diffraction of Light and concept of Zone Plate. 3. Understand the polarization of light.. 4. Study of different types of associated optical instruments based on interference and diffraction of light which are widely used in industry and research.		
Credits: 04		Core Compulsory
Max. Marks: 100 External Exam: 75 Internal assessment: 25		Min. Passing Marks: 33
Total No. of Lectures-Tutorials-Practical (in hours per week): 4-0-0		
Unit	Topic	No. of Lectures
Unit I	Electromagnetic Theory of Light: Definition and Properties of wave front, Plane and Spherical Waves, Longitudinal and Transverse Waves, Maxwell's equations and their interpretations, Poynting's theorem, Energy flux of electromagnetic wave, Huygen's Principle in homogeneous and inhomogeneous medium, Construction of Huygen's wave front.	10
Unit II	Interference: Interference The principle of superposition, Two slit interference, coherence, Division of wave front and amplitude, Optical path retardations lateral shift of fringes, Fresnel biprism, Interference with multiple reflection, Thin films, Application for precision measurements, Haidinger fringes, Fringes of equal thickness and equal inclination.	15
Unit III	Diffraction: Diffraction Fresnel's and Fraunhofer diffraction: Diffraction of single slit, Zone plates, intensity distribution, Resolution of image, Rayleigh criterion, Resolving power of telescopes and microscopes, Diffraction due to 2-slits and N-slits, Diffraction grating, Resolving power of grating and comparison with resolving powers of prisms.	15
Unit IV	Polarizatioin: Polarization Plane polarized, Circular polarized and elliptically polarized light, Malus law, Brewster's law, Double reflection and uniaxial crystals, Application of bi-refringence, Dichroism, Optical rotation, Rotation of plane of polarization, Optical rotation in liquids and crystals, Polarimeter.	10

Unit V	Interferometers: Michelson interferometer and its application for precise measurement of wavelength, Wavelength difference and width of spectral lines, Twyman-Green interferometer, Tolansky fringes, Fabry-Perot interferometer and Etalon.	10
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Suggested Reading

1. D.P. Khandelwaland : Optics and Atomic Physics
2. Jenkins and White : Fundamentals of Optics
3. A.K. Ghatak : Physical Optics
4. Brijlal and Subrahmanyam : Optics
5. K.D. Moltev : Optics
6. B. K. Mathur : Optics

Suggested Online Link:

1. MIT Open Learning - Massachusetts Institute of Technology, <https://openlearning.mit.edu/>
2. National Programme on Technology Enhanced Learning (NPTEL),
<https://www.youtube.com/user/nptelhrd>
3. SwayamPrabha - DTH Channel,
https://www.swayamprabha.gov.in/index.php/program/current_he/8

Suggested Continuous Evaluation (25 Marks):

Continuous internal evaluation shall be based on allotted assignment and class tests. The marks shall be as follows:

Class Test /Assignment (25 marks)

Course Prerequisite: As per the university ordinance.

DEGREE IN SCIENCE		
Programme: <i>Degree in Science</i>		Year: III Semester: V Paper-II
Subject: Physics		
Course Code:	Course Title: Basic Electronics	
Course Outcomes: 1. Study of different Network Theorems for simplifying complicated electronics circuits. 2. Study of Regulated Power Supply. Understand different types of Rectifiers, Filters and Voltage Regulator. 3. Study of different types of special diodes and their applications 4. Study of Bipolar Junction Transistors. 5. Study of Field Effect Transistor		
Credits: 04		Core Compulsory
Max. Marks: 100 External Exam: 75 Internal Exam: 25		Min. Passing Marks: 33
Total No. of Lectures-Tutorials-Practical (in hours per week): 4-0-0		
Unit	Topic	No. of Lectures
Unit I	Network Theorems Superposition Theorem, Constant voltage source and constant current source, Conversion of voltage source into current source, Thevenin's Theorem and procedure for finding thevenin equivalent circuit, Norton's Theorem and procedure for finding Norton equivalent circuit, Maximum power transfer theorem, Applications of Network Theorems	10
Unit II	Power Supplies Semiconductor diode: P-N Junction diode, Diode circuits with DC and AC Voltage sources, Diode as a rectifier: Half and Full wave rectifiers, Bridge rectifiers, Peak inverse voltage, Efficiency, Ripple factor, Filters: Low pass and High pass filters, Band pass and Band stop filters, L and π – filters (Series inductor, Shunt capacitor, LC, CLC filters), Zener diode, its characteristics, Voltage regulation	15
Unit III	Special Diodes Special Diodes Tunneling effect, Tunnel diode, Varactor diode, Point contact diode, V-I characteristic of these diodes, Optoelectronic devices: Light emitting diode (LED), Photo emissive devices, Photodiodes, P-N Junction Photodiodes, PIN photodiode. Avalanche Photodiode	10
Unit IV	Basic Transistor (BJT) Bipolar junction transistor, Transistor operation and its Biasing rule, Transistor currents, Transistor circuit configuration, CB configuration, CE configuration, Relations between α and β , CC configuration, Relations between transistor currents in various configuration. Leakage currents in a Transistor. Transistor	10

	static characteristics in common Base, common Emitter and common Collector configuration, cut-off and saturation points, Active region, h Parameters	
Unit V	Field Effect Transistors (FET) Junction FET, Static Characteristics of JFET, JFET Drain Characteristic with $V_{GS} = 0$, JFET Characteristic with External Bias, Transfer Characteristic, Small Signal JFET Parameters, DC Biasing of a JFET, DC load line, Advantages of FETs, MOSFET or IGFET, Depletion-enhancement (DE) MOSFET, Construction, working and Static Characteristics of a DE MOSFET, Enhancement only N-channel MOSFET, Transfer Characteristics, FETs as Switches, FET Applications, MOSFET Handling.	15

Suggested Reading

1. M.K. Baagde, S.P. Singh and Kamal Singh : Elements of Electronics
2. B.L. Theraja : Basic Electronics
3. V.K. Mehta : Elements of Electronics
4. J.D. Ryder : Networks, Lines and Fields
5. J.D. Ryder : Electronic Fundamentals and Applications.
6. Millman and Halkias : Integrated Electronics

Suggested Online Link:

1. MIT Open Learning - Massachusetts Institute of Technology, <https://openlearning.mit.edu/>
2. National Programme on Technology Enhanced Learning (NPTEL), <https://www.youtube.com/user/nptelhrd>
3. SwayamPrabha - DTH Channel, https://www.swayamprabha.gov.in/index.php/program/current_he/8

Suggested Continuous Evaluation (25 Marks):

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Class Test/ Assignment (25 marks)

Course Prerequisite: As per the university ordinance.

DEGREE IN SCINCE			
Programme: <i>Degree in Science</i>		Year: III	Semester: V Practical
Subject: Physics (Practical)			
Course Code:	Course Title: Demonstrative Aspects of Physical Optics and Demonstrative Aspects of Basic Electronics		
Course Outcomes:			
1. Experimental physics has the most striking impact on the industry wherever the instruments are used to study the Electronics and its application in industry and research.			
2. Measurement precision and perfection is achieved through Lab Experiments.			
Credits: 02		Core Compulsory	
Max. Marks: 50 Internal (Record File): 15 External Practical Exam: 20 External Viva Voce : 15		Min. Passing Marks: 17	
Total No. of Lectures-Tutorials-Practical (in hours per week): 0-0-4			
Unit	Topic	No. of Lectures	
Lab Experiment List			
	1. Biprism- determination of λ . 2. Newton's ring experiment- Determination of λ . 3. Determination of λ by a transmission grating. 4. Zone-plate experiment study of different orders. 5. Malus Law 6. Polarimeter: Specific rotation of sugar solution. 7. To study the characteristics of integrating and differentiating circuit. 8. To draw the characteristics of P-N junction diode. 9. To draw the characteristics of PNP and NPN junction transistor. 10. Measurements of h-parameters of a transistor. 11. Study of different types of Rectifiers and Filters. 12. Verification of Network theorems. 13. Child Langmuir law. 14. Triode/ Tetrode/ Pentode characteristics and constants. 15. Study of power supply (Ripple factor). 16. Study of Zener diode and regulation (taking different source voltage and loads).	60	

	17.To study the Characteristics of a Photo-diode.	
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